CyberEscape: Designing an Educational Escape Room for Cybersecurity Awareness

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Abstract

Traditional cybersecurity awareness training programmes suffer from rapid knowledge decay and low engagement, leaving non-expert users vulnerable to attacks that exploit the human factor. This paper presents CyberEscape, a digital educational escape room (DEER) prototype designed to build cybersecurity awareness through active, scenario-based learning. Grounded in constructivist and situated learning principles, the prototype places players in three realistic threat scenarios: identifying phishing emails in a simulated inbox, evaluating password strength based on publicly visible personal information, and navigating safely on a public Wi-Fi network. The prototype was implemented in Python using the Pygame library, following a sequential checkpoint structure with immediate contextual feedback after each decision. This work demonstrates the feasibility of applying DEER mechanics to cybersecurity education and provides a documented design and working proof-of-concept prototype for further development and evaluation.

CCS Concepts

• **Applied computing** → **E-learning**.

Keywords

Cybersecurity, Digital Educational Escape Rooms, Game-based learning

1 Introduction

The scale of the cybersecurity threat to everyday users and organisations has reached a critical threshold, with threat actors exploiting vulnerabilities at an increasingly rapid pace. According to the ENISA Threat Landscape 2025, phishing remains the dominant intrusion vector, accounting for approximately 60% of observed cases and serving as a frequent precursor to the deployment of malicious code [1]. This landscape is further complicated by a 27% year-on-year increase in newly disclosed vulnerabilities, totalling over 42,000 in the most recent reporting period [1].

While technical security infrastructures continue to advance, cybercriminals have shifted their focus toward exploiting the "human factor," making non-expert users the primary target for attacks like phishing and social engineering [5]. Despite the necessity of effective Security Education, Training, and Awareness (SETA) programmes, traditional approaches remain largely passive, relying on online lectures, videos, and static tutorials that require participants to observe rather than actively learn [5, 6]. These methods can be perceived as uninspiring and tedious by learners, leading to low motivational potential and poor long-term retention of critical security competencies [5]. These methods are frequently perceived as uninspiring and tedious, leading to low motivational potential and rapid knowledge decay, where training increments often evaporate within 15 to 30 days How effective are SETA programmes.

Although game-based learning has emerged as a promising strategy to improve engagement, digital educational escape rooms (DEERs) specifically remain a relatively under-explored area for cybersecurity education [5]. Most existing research has focused on the pedagogical aspects of physical escape rooms, leaving a notable gap in understanding how fully digital, immersive escape environments can be used to foster specific cybersecurity skills [6].

This paper proposes and documents CyberEscape, a digital educational escape room that immerses users in realistic cybersecurity scenarios to build awareness through active decision-making and immediate feedback. By placing learners in a "safe" virtual environment where they can experience the immediate consequences of their choices.

The remainder of this paper is structured as follows. Section 2 reviews related work on game-based learning and cybersecurity education. Section 3 analyses the cyberattack landscape and justifies the selected threat scenarios. Section 4 presents the design of the escape room. Section 5 describes the prototype implementation. Section 6 concludes the paper.

2 Related work

2.1 Game-based learning

Game-based learning (GBL) refers to the use of games as instructional tools where learning objectives are inherently integrated into the game structure [5, 9, 10]. Research has consistently shown that GBL produces superior learning outcomes compared to conventional teaching methods; for example, students in a GBL group achieved significantly higher learning gains than a control group, with a statistical significance of $p<0.05$ [16]. A core advantage of this approach is the concept of "graceful failure," where the game acts as a "safe space" that treats mistakes as necessary, low-consequence steps in the learning process rather than final negative outcomes [5, 9, 10]. This environment fosters the inner drive of a player, or intrinsic motivation [9, 10]. When applied as educational escape rooms, GBL leverages high situational interest to engage participants in active problem-solving and decision-making under time pressure [6]. These characteristics transform learners from passive receivers of information into active participants, making the format especially relevant for cybersecurity, where users must practice recognising and responding to realistic threats in a risk-free environment [6],for example, emotions like urgency or fear might be replicated by the time constraints of an escape room.

2.2 Escape Rooms in Education

An Educational Escape Room (EER) is a game, generally team-based, filled with puzzles to be solved in a determinate amount of time [17] with the goal of gaining knowledge in a subject or even developing domain specific skills [14] especially in STEM subjects like Computer Science, physics, mathematics among others [6]. They are aimed at a specific target group and have clear learning goals while aligning with a curriculum [14].EER have a high success rate outcome, avoiding frustration and boredom, while giving students a positive learning experience and keeping them engaged [4, 6, 14]. In order to win, one must solve all the challenges before the time runs out to escape the room, although recently the end can also be opening a vault or solving a crime [14]. Some important design principles are the narrative of the game to put the knowledge and skills needed in context, the immersion of the players to get learners engaged, consistency of the time in which the EER takes place, so no jumping around completely different places and time, and also the decorations and props are recommended to not create a cognitive dissonance in learners [14]. There can be different types of puzzles, cognitive puzzles that require thinking and logic, physical puzzles that require moving objects or opening doors and vaults, and meta-puzzles which are the last puzzle in the Escape Room (ER) derived from all the previous puzzles to give an end and resolution to the ER [14]. There are 4 main ways to design the puzzle structure. There is the open structure where we have independent puzzles that can be solved in any sequence, but all needed to the meta-puzzle. There is the sequential design, which is a sequence of puzzles in which to get to the next puzzle you have to solve the one before. Then there is the path-based structure in which there are different paths, each path is one sequential puzzle, and each path has to be solved to get to the meta-puzzle. Lastly, there is the pyramid, which is a mix of everything. There are also 2 main formats for ERR, there are physical and digital EERs (DEER). They both have the same goals, mainly the acquisition of knowledge and content-related skills, behavioural/social skills principally for team-based EERs, and affective skills such as motivation, engagement, and satisfaction [6]. There have been no significant differences between these two formats which suggests that they are both similarly effective [5]. Although these learning goals, EERs should be used to enhance or improve learning, such as digital skills for DEERs, not to replace courses. Finally, Makri et al. [6] found that DEERs results have been positive, increasing engagement, motivation, and learning results, as well as increased academic performance. It gave students a deeper understanding of new concepts in various subjects, among them diabetes, programming, mathematics, and numerous others.

2.3 Cyber Security Awareness Education

There exist cybersecurity awareness training programmes such as security education, training and awareness (SETA) programmes, but in these SETA programmes cybersecurity knowledge usually increases by 12-17% and then disappears within a month [8]. These SETA programmes, which are predominant in the professional field, are often seen as tedious and uninspiring by the people who take them [5]. Cybersecurity awareness is extremely important, as human factors, such as social engineering tactics, remain the primary point of entry for threat actors, with 60% coming from phishing [1].

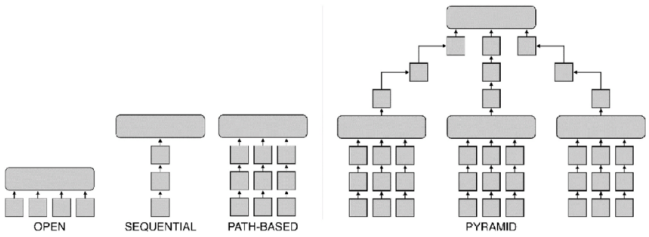


Figure 1: Figure 1: Puzzle structures in escape rooms: a) basic structures: open, sequential and path-based; b) a complex, hybrid structure, such as a pyramid. Squares are puzzles and rectangles are meta-puzzles (adapted from Nicholson, 2015) as cited in [14].

This is also supported by Verizon's 2025 Data Breach Investigation Report (DBIR) which found that 60% of breaches consist of human involvement [15]. In addition, 66% of Chief Information Security Officers (CISO) agree that the human factor is their organisation's biggest cyber vulnerability, making it clear that organisations are aware of risks [11]. The intersection of DEERs and cybersecurity awareness education for non-expert users is under-explored. Most existing EERs target general STEM subjects or medical education, but not necessarily cybersecurity-specific implementations [5, 6]. There have been some DEERs for cybersecurity like EyesOnCS by Spatafora et al., although they had some deficiencies, for example immersion challenges, the difficulty being too high, and even their hint system [12]. So, there are some DEERs, but they usually lack evidence of long-term knowledge and act as short-term interventions [3]. Additionally, there is not much evidence for actual behavioural change, which is the goal here.

3 Cyberattack Landscape Analysis

The selection of threat scenarios in CyberEscape involved a review of the ENISA Threat Landscape 2025 and the 2025 Verizon Data Breach Investigation Report. These reports were analysed to identify attack vectors that impact non-expert users and organisations where the "human factor" is the primary entry element [1, 15]. The selected scenarios, phishing, password-based attacks, and unsecured network connections, were prioritised because they do not require special and technical knowledge for a user to fall victim to, yet they can be avoided by behavioural training and scenario-based decision-making [5, 13].

3.1 Phishing and Social Engineering

Phishing is defined as a fraudulent attempt to trick people into revealing sensitive information, such as login credentials or financial data, by masquerading as a trustworthy entity in electronic communications, usually email or SMS texts [2]. It is the dominant intrusion vector in the current landscape, accounting for 60% of all observed cases and 74% of phishing sites even using HTTPS to seem like a secure website [1, 2]. Phishing targets non-experts by exploiting human nature, using psychological triggers such as urgency, fear, or even authority to provoke an impulsive and emotional response that bypasses rational evaluation [2, 5]. These psychological manipulation tactics used on people to make them act and divulge

sensitive information are known as social engineering. A prevalent type of phishing is "spear phishing" which relies on research on the victims to make the scam appear more authentic leading to one of the mist successful types of attacks [2]. To avoid these traps, users must be taught to identify specific red flags: suspicious sender addresses, inconsistencies in URLs/hyperlinks, and requests for sensitive information that deviate from professional standards [2]. In this project, this translates into a puzzle mechanic where the player is presented with a simulated email inbox with phishing emails that the player must identify with help of the red flags.

3.2 Password-based attacks

Credential attacks involve attempts to obtain unauthorised access to systems by exploiting user passwords through methods such as brute force, the act of systematically guessing credentials, password guessing, and credential stuffing; the act of using leaked passwords from one service to access another [12, 15]. These actions are spreading, with credential abuse accounting for 32% of human element breaches [15]. Non-experts remain vulnerable because we humans are historically bad at choosing unfamiliar and complex passwords and frequently reuse them across multiple platforms [12, 15]. The teachable elements of this scenario focus on strong passwords such as using long passphrases and mnemonics, the necessity of having unique passwords across platforms, and the role of password managers [13]. This becomes a puzzle mechanic where players must evaluate a list of credentials to identify weak or reused ones, or must generate a secure password to complete the puzzle.

3.3 Network Sniffing and Unsecured Connections

Network sniffing is a technique in which an attack on the same network passively captures data traffic. This occurs most frequently on public Wi-Fi and/or unsecured networks where unencrypted credentials, session keys, and personal information are exposed. According to ENISA, the network server is the attack vector for 64% of documented vulnerabilities [1]. Teachable elements for non-experts include recognising the difference between HTTPS and HTTP, the protective roles of Virtual Private Networks (VPNs), the risks of "shadow" connections. In an ER context, this is presented as a scenario where the player must analyse a virtual workplace environment and identify which networks connections are safe by verifying the "lock" icon in a browser, among other possibilities, and which expose data to potential interception.

4 Escape Room Design

4.1 Pedagogical Framework

The design of CyberEscape is grounded in three core pedagogical pillars: **constructivism**, **situated learning**, and **immediate feedback**.

Firstly, the game uses a constructivism approach, which assumes that learners are not passive recipients of the information being taught, but active participants who construct new knowledge through real-time experiences and interaction with their environment [6, 9, 10]. In CyberEscape, this is achieved by requiring players to solve puzzles requiring understanding of the information given, as well as using it in a real scenario.

Secondly, the design leverages situated learning theory, which argues that knowledge is best acquired in the authentic context where it will later be applied [9, 10, 14]. By embedding cybersecurity challenges within a realistic workplace scenario, such as an email inbox, or a public Wi-Fi setting, the game provides a safe environment for learners to rehearse skills and practice decision-making without the real-world consequences like giving away their data, passwords, or other sensitive information [7, 12]. This scenario-based learning facilitates better knowledge retention and the immediate transfer of skills to the professional environment, which is why immersion is very important, making the learner both realise where these skills and information can be used, keeping them engaged, motivated and if they solve the puzzles even give them a sense of achievement, in conclusion giving them an active learning environment maximising motivation and engagement [6].

Lastly, immediate and unambiguous feedback is a critical design element [14]. The game uses a loop of challenge, response, and feedback, where incorrect decisions are immediately explained to the player, thus correcting him/her when making the mistake instead of waiting until the end when he/she might have already forgotten the mistake. This turns failure into a "learning moment", allowing the player to understand exactly what they missed or did wrong, and reinforcing correct identification strategies before they move on to the next puzzle or challenge.

Together, these three principles ensure that every design choice prioritises active engagement over passive consumption. The game will be single-player, making it lose the communication skills reinforcement, but therefore I focused more strongly on other principles.

4.2 Learning Objectives

The following table outlines the specific learning objectives targeted by the threat scenarios in CyberEscape.

Scenario Name	Puzzle Name	Learning Objective
Phishing	The Deceptive Inbox	Player can identify at least 3 red flags in a malicious email.
Password Security	The Profile Problem	Player understands what distinguishes a weak from a strong password and why.
Network Sniffing	The Café Connection	Player can identify unsafe connection behaviour on a public network.

Table 1: Learning Objectives for CyberEscape Scenarios

4.3 Narrative and Room Structure

CyberEscape places the player in everyday situations that anyone who uses the internet will recognise. As this is a proof-of-concept prototype, narrative immersion is kept deliberately light. The priority is the learning outcome that players take from this DEER, as well as the puzzle mechanics. The player will navigate three

sequential security checkpoints, each set in a different context from daily (digital) life. There is no previous technical background assumed or required to complete these puzzles. They are aimed at every age group that has a brief understanding of the internet and applications such as email inboxes.

The game follows a sequential structure [14]: each checkpoint must be completed before the next one becomes accessible, ensuring that the three threat scenarios are encountered in a deliberate order of increasing contextual variety. Each checkpoint opens with a short screen to set the scene, thus establishing the situation without telling the player what red flags to look for until after the completion of the puzzle when the red flags will finally be revealed. The discovery is therefore the learning. Upon completing the puzzle, the player receives an explanatory debrief that consolidates the lesson regardless of the outcome. This means that a player who completes every puzzle in 10 minutes and a player who completes it in 60 minutes or even not at all will both get an explanation as to what the red flags were and are, and what they should look out for, ensuring that every player learns something.

Figure 2 illustrates the overall progression structure of CyberEscape.

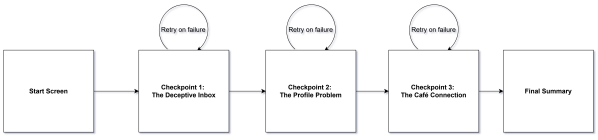


Figure 2: CyberEscape sequential progression flow. Each checkpoint must be cleared before the next is accessible. A debrief screen follows every attempt, regardless of outcome.

Table 2 summarises the three checkpoints, their corresponding threat scenario, and their real-world setting.

No.	Checkpoint	Threat Scenario	Setting
1	The Deceptive Inbox	Phishing & Social Engineering	Personal or work email client
2	The Profile Problem	Password-Based Attacks	Social media profile and login screen
3	The Café Connection	Network Sniffing	Café during a break

Table 2: CyberEscape checkpoint structure and sequential progression order.

4.4 Puzzle Designs

4.4.1 Checkpoint 1: The Deceptive Inbox. The player is presented with a simulated email client split into two panels. The left panel shows five email previews, each displaying the sender address and subject line. Clicking a preview opens the full email in the right panel, showing the From, To, Subject, and message body fields. The email pool is drawn from a larger data set containing four legitimate emails and two phishing emails; each session selects three legitimate emails at random and always includes both phishing emails, giving five in total.

The player must classify each email as **PHISHING** or **SAFE** using two buttons at the bottom of the viewer. Once all five are labelled, pressing the Enter key submits the answers. Both phishing emails contain identifiable red flags grounded in real attack patterns [2]. The first originates from **support@rnicrosoft.com**, where the character combination "rn" visually mimics the letter "m", a technique known as typosquatting. Its subject threatens account suspension within 24 hours and asks the player to click a verification link. The second arrives from **security@ing-lU.com**, falsely claiming a suspicious transaction and demanding verification via a link that does not match any official domain.

Any misclassification triggers a three-second retry screen and resets the puzzle with a freshly shuffled email set. Players who classify all five correctly advance to the debrief screen, which consolidates the key lessons: how to read sender addresses character by character, how urgency is used as a manipulation tactic, and why no legitimate service will ever request credentials through an emailed link [2].

4.4.2 Checkpoint 2: The Profile Problem. The screen is divided into two panels. The left panel displays a fictional social media profile belonging to Sarah Mitchell (**@sarah_m_lux**), showing her birthday, location, dog's name, favourite football club, and favourite player, alongside recent posts that repeat the same details. The right panel hosts the interactive puzzle, which runs in two sequential phases.

In Part A, the player is shown six password candidates, three derived from Sarah's profile and three not, and must identify which one she is most likely to be using. In Part B, the player is shown four candidates and must identify the genuinely strong one. Each phase requires the player to confirm their selection before receiving feedback.

In Part A, any password that draws directly from the visible profile information is accepted as correct; examples include **Biscuit2022**, **Chelsea1990**, or **Sarah_14March**. The point is not a single right answer, but that the player demonstrates understanding of why personal information makes passwords predictable [12, 15]. In Part B, the only correct answer is **Lamp!River93Desk**, a passphrase of three unrelated words combined with a number and a symbol. Selecting a non-personal password in Part A or a weak one in Part B resets that phase and shows a brief explanation before the player tries again. The debrief screen after the checkpoint explains the connection between oversharing and credential vulnerability, the properties of a strong passphrase, and the role of password managers [13].

4.4.3 Checkpoint 3: The Café Connection. The player is told they are on a break at a café and need to quickly check something online. The checkpoint runs across three sequential steps, each with its own instruction and a set of selectable options.

Step 1 presents four Wi-Fi networks. Three are unsafe: an open network with no password, a network labelled “Guest” that is also unencrypted, and an unidentified personal hotspot. The correct choice is the single WPA2 password-protected network. Selecting an unsafe option shows an explanation specific to that network and resets the selection; the player cannot advance until the correct network is chosen. Step 2 displays two simulated browser address bars. One shows **http://mybank.lu/login** with a warning indicator; the other shows **https://mybank.lu/login** with a padlock "icon". Again, an incorrect selection shows an explanation and resets before the player can advance. Step 3 presents six everyday online actions and asks the player to mark each as **SAFE** or **RISKY**. Reading the news or checking the weather are safe; logging into online banking, entering credit card details, or sending work email containing private content are risky even on a nominally protected network.

Step 3 is the only phase where submitting an incorrect set of answers triggers a full retry. A module-level flag ensures that the retry restarts from Step 3 only, preserving the player’s progress through Steps 1 and 2. The debrief consolidates the three key takeaways: how open networks expose unencrypted traffic, what the HTTPS padlock guarantees, and which actions should be avoided on any public connection [1].

5 Prototype Implementation

5.1 Technology Stack

CyberEscape is implemented entirely in Python, chosen for its readability and the availability of the Pygame library. The prototype runs on Python 3.13 and depends on a single third-party library, Pygame 2.6.1, which provides hardware-accelerated 2D rendering, event handling, font rendering, and a fixed-rate game loop. No other external dependencies are required. I have also chosen PyGame because of my experience with it in high school, where Python and PyGame were my main programme for my Computer Science lectures.

The game launches in fullscreen mode and reads the display resolution at startup, computing all layout dimensions dynamically so that the interface scales to any screen. All puzzle content, including email text, password candidates, network configurations, and action labels, is stored as plain Python data structures in a dedicated constants module. This separation of content from rendering logic means that puzzle data can be modified or extended without touching the drawing code.

5.2 System Architecture

The application is built around a flat state machine coordinated by a central game loop in `main.py`. At any given moment the application is in exactly one of eleven named states:

`intro`, `puzzle1_intro`, `puzzle1`, `puzzle2_intro`, `puzzle2`, `puzzle3_intro`, `puzzle3`, `summary1`, `summary2`, `summary3`, and `retry`. State transitions are triggered by the return value of the active module’s `handle()` function.

Each module, `Menu.py`, `puzzle1.py`, `puzzle2.py`, `puzzle3.py`, and `checkpoints.py`, exposes three module-level functions with a uniform interface: `setup()` initialises or resets the module’s state; `handle(event)` processes a single Pygame event and returns either `None` (no transition) or a string identifying the next state; and `draw(screen)` renders the current frame. This keeps the main loop free of module-specific logic.

A shared utility module, `txt_rect.py`, provides two dynamic text-rendering functions that wrap a string to fit inside a given rectangle and scale the font size downward until the full text fits, removing the need to manually calibrate font sizes for different screen resolutions. A 15-minute countdown timer is maintained in the main loop; when it expires, the game advances directly to the summary screens regardless of the player’s current progress, ensuring every player receives the full educational debrief.

5.3 Implemented Mechanics

Puzzle 1 implements a two-panel email client simulation. Each email is represented as an object carrying sender, recipient, subject, body text, and a `flagged` attribute (True, False, or None if unclassified). The player assigns each email a classification using PHISHING or SAFE buttons; submission is blocked until all five carry a label. Pressing Enter evaluates each classification against its ground-truth value and either advances the game or triggers a retry with a reshuffled email set.

Puzzle 2 implements a two-phase password evaluation mechanic alongside a persistent social media profile panel. In Phase A, six password candidates are displayed as selectable buttons and the player must identify one derivable from the visible profile information. In Phase B, four candidates are shown and the player must select the strongest one. Each phase requires an explicit confirmation click before feedback is shown; incorrect selections reset only the current phase, allowing the player to retry after reading the explanation.

Puzzle 3 is structured as three sequential micro-tasks within a single checkpoint. Phase 1 uses a card-based network selector with one option per network; Phase 2 simulates two browser address bars with colour-coded security indicators; Phase 3 uses a row-based interface where each of the six action rows has dedicated SAFE and RISKY buttons. A module-level variable (`retry_phase`) records which phase failed, so that a Phase 3 retry restarts only from Phase 3 rather than from the beginning of the checkpoint.

All three puzzles share the same interaction pattern: select an option, confirm, receive immediate feedback, and either advance or correct the answer. Post-checkpoint debrief screens rendered by the `checkpoints` module consolidate the educational content for each threat scenario, and are shown to every player regardless of performance.

6 Conclusion

This paper has presented CyberEscape, a digital educational escape room prototype designed to teach foundational cybersecurity awareness to non-expert users through active, scenario-based decision-making. Grounded in constructivist and situated learning theory, the prototype delivers three puzzle-based checkpoints covering phishing identification, password security, and safe behaviour on public Wi-Fi networks. Each checkpoint provides immediate contextual feedback and concludes with an explanatory debrief

screen, ensuring that every player receives the educational content regardless of their in-game performance.

The implementation demonstrates that Python and Pygame provide a sufficient foundation for a functional and modular DEER prototype, with dynamic content randomisation, resolution-independent layout, and a clean separation between game content and rendering logic.

Several limitations remain. The prototype has not been evaluated with real users, so claims about learning effectiveness are hypothetical at this stage. The single-player format, while appropriate for a proof-of-concept, removes the social interaction and collaborative problem-solving that characterise physical escape rooms and that some research associates with stronger engagement and retention [4, 14]. Narrative immersion is intentionally minimal and could be substantially deepened in a more complete implementation. The current puzzle set also covers only three of the many cybersecurity threat scenarios relevant to non-expert users.

Future work should prioritise a formal user evaluation with a representative sample to assess both perceived engagement and actual learning outcomes. Extending the puzzle set to additional scenarios, such as social media oversharing or two-factor authentication, would broaden the educational scope. A multiplayer mode would allow investigation of whether collaborative play produces stronger outcomes than the current single-player format, in line with findings from the broader educational escape room literature [6, 14].

A Disclosure of Generative AI Use

In accordance with the University of Luxembourg Guidelines on the Use of Generative AI Tools for Learning and Teaching (2025), I want to disclose the following use of AI tools in the preparation of this report.

Claude (Anthropic)

Claude was used for two distinct purposes. First, it provided coding assistance during the implementation of the prototype. Rather than generating the application as a whole, Claude was primarily consulted for guidance on how to implement specific features or how to structure certain components (for example, the implementation of individual puzzles or aspects of the programme architecture). The prototype itself was written by me, including the design choices and the overall behaviour of the application. Claude's suggestions were selectively incorporated where appropriate, while all code was reviewed, tested, and adapted manually by me.

Second, Claude was consulted during the structural planning of this report, specifically for guidance on how to organise the sections and subsections in accordance with academic writing conventions. The intellectual content of each section, including the literature review, the threat scenarios selected, the design decisions made, and the arguments presented, is my own work. Claude did not generate any factual claims, citations, or analysis included in this report.

NotebookLM (Google)

NotebookLM was used as a source management and summarisation tool during the literature review phase. Uploaded research papers were queried to assist in organising and cross-referencing sources, and the tool was occasionally used to help rephrase passages for clarity while preserving factual accuracy. All rephrased content was reviewed and verified against the original sources by me, before including it in the report.

Impact on the Work

The use of these tools affected the presentation and organisation of this work, but not its substance. The research questions, the selection of threat scenarios, the pedagogical design choices, and the conclusions drawn are the result of my own reading, thinking, and decision-making, as well as the help of my supervisor Professor Muriel-Larissa FRANK. AI assistance was used to improve clarity, accelerate structuring, and support technical implementation, not to replace the intellectual work required by the project.

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